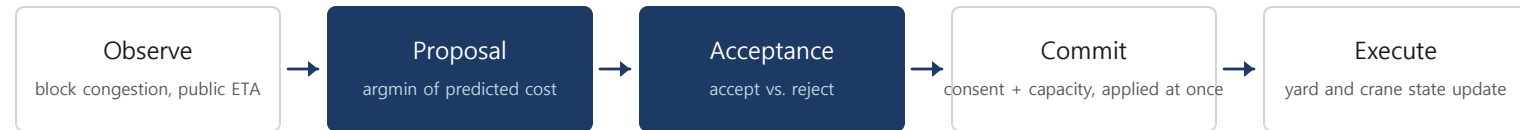


Fig. 1. Reallocation decision flow and the learning/operation split

Operating path — trained networks only, zero counterfactual calls



Learning path — clone state and random stream, run up to three worlds for 3 h
observed / proposal reversed / acceptance reversed → centred Φ difference as target

Only the cost difference produced in training enters the policy; operation invokes no branch simulation.